

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML-C-13
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI Digital Health Twin

Abstract

Healthcare systems across the world primarily follow a reactive approach, where diseases are diagnosed and treated only after noticeable symptoms appear. Although wearable devices and Internet of Medical Things (IoMT) sensors continuously collect physiological parameters such as heart rate, blood pressure, oxygen saturation (SpO₂), glucose levels, sleep quality, and physical activity, most existing healthcare applications merely display raw health data or provide simple threshold-based alerts. These systems lack intelligent prediction, personalized simulation, long-term health modeling, and preventive recommendations. As a result, patients, especially those suffering from chronic diseases, elderly individuals, and people living in remote areas, often experience delayed diagnosis and increased healthcare costs.

Several recent research works have focused on AI-assisted healthcare monitoring using machine learning algorithms, IoMT devices, cloud computing, and predictive analytics. Existing Digital Health Twin models mainly concentrate on disease prediction or remote patient monitoring independently. Most available solutions fail to integrate real-time IoMT data, personalized digital twin simulation, explainable AI, lifestyle recommendations, emergency alert mechanisms, and continuous health trajectory prediction into a single unified platform. Furthermore, many previous models provide prediction without enabling users to visualize how lifestyle modifications such as improved diet, exercise, medication adherence, and sleep patterns may influence future health outcomes.

To address these research gaps, this project proposes an AI-Based Digital Health Twin for Predictive Healthcare Using IoMT and Machine Learning. The proposed system creates a personalized virtual replica (Digital Twin) of every patient's health by continuously collecting physiological data from IoMT devices such as smartwatches, pulse oximeters, ECG sensors, glucometers, and blood pressure monitors. Advanced machine learning algorithms including XGBoost and Scikit-learn analyze both historical and real-time health data to predict potential risks related to cardiovascular diseases, diabetes, hypertension, kidney disorders, sleep abnormalities, and stress.

The proposed implementation utilizes Flutter for mobile application development, Python with FastAPI for backend services, MongoDB Atlas for cloud database management, Firebase Authentication and Cloud Messaging for secure communication and notifications, and AI/ML libraries such as Scikit-learn, XGBoost, Pandas, and NumPy for predictive analytics. The required hardware includes IoMT-enabled wearable devices, smartphones, cloud servers, internet connectivity, and optional medical sensors such as ECG modules, glucometers, pulse oximeters, and blood pressure monitors.

Through an extensive literature survey of recent Digital Twin, Artificial Intelligence, IoMT, and predictive healthcare research published between 2023 and 2026, this work aims to develop an intelligent preventive healthcare platform capable of improving early disease detection, reducing unnecessary hospital visits, supporting continuous remote patient monitoring, and promoting personalized healthcare through predictive simulation and data-driven clinical decision support.

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References (Category specific)

1. Kabir, M. R., et al. (2025).
"Digital Twins in Healthcare IoT: A Systematic Review."
Smart Health, Elsevier.
DOI: <https://doi.org/10.1016/j.smhl.2025.100716>

2. Rudsari, H. K., et al. (2025).
"Digital Twins in Healthcare: A Comprehensive Review and Future Directions."
Scientific Reports (Nature Portfolio).

3. Lukaniszyn, M., et al. (2024).
"Digital Twins Generated by Artificial Intelligence in Healthcare."
Applied Sciences, MDPI, Vol. 14(20).

4. Vallée, A., et al. (2023).
"Digital Twin for Healthcare Systems."
Frontiers in Digital Health.

5. Chen, J., Shi, Y., Yi, C., Kang, J., & Niyato, D. (2024).
"Generative AI-Driven Human Digital Twin in IoT-Healthcare: A Comprehensive Survey."
arXiv Preprint.

6. Otapo, A. T., Othmani, A., Khodabandelou, G., & Ming, Z. (2024).
"Prediction and Detection of Terminal Diseases Using Internet of Medical Things: A Review."
arXiv Preprint.

7. Developing Medical Digital Twins through Omics, IoT, and Artificial Intelligence for Preemptive Medicine (2024).
Journal of the Japanese Medical Association (JMA Journal).

Project Team Members

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Project Guide

Name: **Ms.M Baby Anusha**

Signature:

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INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.